WEB PROGRAMMING PROJECT

E-Commerce Platform for Wine Store

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1. Introduction

This project is a web-based E-commerce platform developed for a wine store (Enoteca). The main objective of the project is to provide customers with an online environment where they can browse products, view details, add products to a shopping cart, and place orders.

The platform also includes an administrator area that allows product management through Create, Read, Update and Delete (CRUD) operations.

The project has been developed using PHP, MySQL, HTML, CSS and Bootstrap and follows the requirements specified in the course guidelines.

2. Motivation and Objectives

The motivation behind this project is the increasing importance of online commerce in modern business environments.

The objectives of the project are:

• Create a complete E-commerce platform.
• Implement user registration and authentication.
• Store data using a relational SQL database.
• Provide an administrator dashboard.
• Allow customers to browse products and place orders.
• Apply web programming concepts learned during the course.

The project simulates a real-world online wine store and demonstrates the integration between frontend and backend technologies.

3. Technologies Used

The following technologies were used during the development process:

Frontend:
- HTML5
- CSS3
- Bootstrap 5

Backend:
- PHP

Database:
- MySQL (MariaDB)

Development Environment:
- XAMPP
- Visual Studio Code

Version Control:
- Git
- GitHub

These technologies were selected because they are widely used in modern web development and allow rapid implementation of database-driven web applications.

4. Database Design

The database was designed using MySQL and contains the following tables:

Users
Stores registered users and administrators.

Products
Stores wine information such as name, description, price and image.

Orders
Stores customer orders and total purchase amount.

Order_Items
Stores products associated with each order.

Relationships:

• One user can create multiple orders.
• One order can contain multiple products.
• Products are linked to orders through the Order_Items table.

This database structure ensures data consistency and scalability.

5. Implemented Features

User Features:

• User Registration
• User Login
• User Logout
• Product Browsing
• Product Details Page
• Shopping Cart
• Checkout Process

Administrator Features:

• Administrator Login
• Product Creation
• Product Editing
• Product Deletion
• Product Management Dashboard

The system uses session management to maintain user authentication and shopping cart information.

6. Development Decisions

Several design decisions were made during the development process.

PHP was selected as the backend language because it integrates easily with MySQL and is supported by XAMPP.

Bootstrap was used to improve responsiveness and user interface consistency.

The project follows a modular folder structure separating authentication, administration, configuration and public pages.

GitHub was used to manage source code and version control.

The visual design was adapted to represent a professional wine store environment.

7. Conclusion

This project successfully implements a complete E-commerce platform for a wine store using PHP and MySQL.

The system provides user authentication, product management, shopping cart functionality and order processing features.

The project demonstrates the practical application of web programming concepts including frontend development, backend development, database management and version control.

Future improvements may include online payment integration, product search, order tracking and advanced reporting functionalities.